

# Matrices and Determinants

Rajat Kalia

19 August, 2026

## Infrastructure

#764, 41-A, Chandigarh {TC(official)} - 160036  
 Shamlat (Ramgarh Sardaran), Ludhiana, Punjab  
 SCO 24(TF), 41-D, Chandigarh - 160036  
<https://www.alphaclasses.com/>

## Educational

<https://github.com/rajatkalia1982>  
<https://sourceforge.net/u/rajatkalia/profile>  
<https://www.youtube.com/@rajatkaliaeducation>

## Store

<https://www.lulu.com/spotlight/rajatkalia11>  
<https://amazon.com/author/rajatkalia>

## Sports

<https://www.youtube.com/@indiangamersorganization>  
<https://www.facebook.com/groups/igohn>  
<https://india.voobly.com/>

## Social

<https://www.goodreads.com/sardarbhagatsingh>  
<https://x.com/RajatKalia1>  
<https://www.linkedin.com/in/rajatkalia007/>  
<https://www.facebook.com/rajatkalia007>  
<https://www.instagram.com/rajatkalia007/>  
<https://www.twitch.tv/rajatkalia>  
<https://www.facebook.com/rajatkaliaeducation>  
<https://www.researchgate.net/profile/Rajat-Kalia>  
<https://www.w3profile.com/rajatkalia/>  
<https://www.reddit.com/user/rajatkalia/>

## Blogs

<https://rk-e.blogspot.in/> [Physics]  
<https://chem-rke.blogspot.com/> [Chemistry]  
<https://biosciences-rke.blogspot.com/> [ Bio-Sciences ]  
<https://rk-edu.blogspot.com/> [ Mathematics ]  
<https://medlife-rke.blogspot.com/> [Medical Sciences ]  
<https://computers-mke.blogspot.com/> [ Computers Engineering ]  
<https://ee-mke.blogspot.com/> [Electrical Engineering ]  
<https://me-mke.blogspot.com/> [ Mechanical Engineering ]  
<https://ce-mke.blogspot.com/> [ Civil Engineering ]  
<https://mke-whj.blogspot.com/> [ White Hat Junior ]  
<https://ba-xpn.blogspot.com/> [ Bachelor of Arts ]  
<https://constitution1982.blogspot.com/> [ Constitution ]  
<https://debk-mke.blogspot.com/> [ Accounting ]  
<https://comm-pke.blogspot.com/> [ Commerce ]  
<https://brigade-13.blogspot.com/> [ Air Force, India ]  
<https://battalion-7.blogspot.com/> [ Armed Forces ]  
<https://design-mke.blogspot.com/> [ Design ]  
<https://tmm18.blogspot.com/> [ Meditation ]

## Freelancing

<https://www.guru.com/freelancers/rajat-kalia>  
<https://www.codecademy.com/profiles/rajatkalia>  
<https://leetcode.com/u/rajatkalia1/>  
<https://www.scribd.com/user/308680256/Rajat-Kalia>

Part I

Matrices Problem Set

# Chapter 1

## JEE Mains Problems

Q1 : Let  $A = \begin{bmatrix} 1 & 2 \\ 1 & \alpha \end{bmatrix}$  and  $B = \begin{bmatrix} 3 & 3 \\ \beta & 2 \end{bmatrix}$ . If  $A^2 - 4A + I = 0$  and  $B^2 - 5B - 6I = 0$ , then among the two statements :

$$(S1) : [(B - A)(B + A)]^T = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$$

$$(S2) : \det(\operatorname{adj}(A + B)) = -5,$$

(A) Only (S1) is correct

(B) Only (S2) is correct

(C) Both (S1) and (S2) are correct

(D) Both (S1) and (S2) are wrong

{Solution : <https://www.youtube.com/watch?v=uTUbsovqU8>}